Advances in Explainable Artificial Intelligence Since 2019: Methods, Applications, and Emerging Challenges

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Abstract

The field of Explainable Artificial Intelligence (XAI) has undergone rapid transformation since 2019. Driven by widespread deployment of deep learning in high-stakes domains, growing regulatory pressure, and the emergence of large-scale foundation models, the research community has produced new techniques, evaluation frameworks, and theoretical insights. This survey synthesises key advances across four themes: (1) post-hoc and ante-hoc explanation methods, including counterfactual generation, concept-based approaches, and prototype learning; (2) explainability for attention-based architectures and mechanistic interpretability for large language models (LLMs); (3) XAI for graph neural networks (GNNs); and (4) evaluation methodology. We also survey domain applications and close with open problems including adversarial robustness of explanations, privacy tensions, and the regulatory landscape of the EU AI Act.

Keywords: Explainable AI, Interpretability, Deep Learning, Transformers, Large Language Models, Graph Neural Networks, Evaluation, Counterfactual Explanations, Mechanistic Interpretability, EU AI Act

1 Introduction

The opacity of modern machine learning models has long been a source of concern. Deep neural networks, despite their remarkable predictive power, arrive at decisions through computations not directly interpretable by humans. This tension is consequential: a credit-scoring model that cannot explain a rejection may violate consumer-protection law; a medical diagnostic system whose reasoning cannot be audited may never gain clinical adoption; an autonomous vehicle that cannot account for an evasive manoeuvre cannot be certified as safe.

Explainable Artificial Intelligence (XAI) encompasses the techniques, frameworks, and evaluation criteria that seek to make AI behaviour understandable to human stakeholders. The field matured substantially between 2016 and 2018 with landmark contributions including LIME [1], SHAP [2], Integrated Gradients [3], Grad-CAM [4], and the theoretical foundations of Doshi-Velez and Kim [5] and Lipton [6]. The 2019 survey by Arrieta et al. [7] marked a consolidation point for that first generation. Rudin's influential call to abandon black-box models in high-stakes settings in favour of inherently interpretable models [8] also appeared in 2019 and framed much subsequent debate.

Since 2019, the landscape has shifted in four important ways. **First**, Transformers and their descendants (BERT, GPT, ViT) have replaced CNNs and RNNs as the de facto standard, requiring new explainability strategies. **Second**, foundation models pre-trained on internet-scale corpora have blurred the boundary between explanation target and explanation medium: an LLM can itself produce natural-language explanations. **Third**, the EU AI Act [9] mandates transparency requirements for high-risk applications, converting XAI from a research aspiration to a legal obligation. **Fourth**, evaluation methodology has received sustained critical attention, clarifying what "good" explanations look like.

This survey covers these developments from 2019 through early 2025, identifying the most influential conceptual threads rather than attempting encyclopaedic coverage.

2 Advances in Post-Hoc Explanation Methods

2.1 Feature Attribution: Axiomatic Consolidation

The period 2019–2021 saw significant theoretical consolidation of feature attribution. Sundararajan and Najmi [10] showed that SHAP, Integrated Gradients, and several other methods are instances of a more general Shapley-value framework parameterised by a choice of *baseline*, clarifying previously confusing discrepancies between methods.

Lundberg et al. [11] introduced *TreeSHAP*, which computes exact Shapley values for tree ensembles in polynomial time. Widely used in production systems, it enabled SHAP to become the dominant explanation framework for tabular data, and introduced the important distinction between *path-dependent* and *interventional* SHAP.

2.2 Counterfactual Explanations

Counterfactual explanations — answers to "what would need to change for the decision to be different?" — emerged as a practically compelling paradigm. Building on Wachter et al. [12], the post-2019 literature focused on counterfactuals that are simultaneously *actionable*, *sparse*, *plausible*, and *diverse*. The DiCE framework [13] addressed diversity directly by formulating generation as an optimisation problem with an explicit diversity penalty. A notable thread concerns *causal counterfactuals*, which ground changes in structural causal models to ensure proposals respect the causal structure of the world [14].

2.3 Concept-Based and Prototype Explanations

Feature attribution methods identify which inputs mattered but not *why* at a semantic level. Concept-based methods bridge this gap. **ACE** [15] extended TCAV [16] by automatically discovering concepts through multi-resolution image segmentation and activation-space clustering, removing the requirement for manually labelled concept examples. **ProtoPNet** [17] built interpretability into the architecture itself: the network learns prototypical parts used as comparison templates, explaining predictions as "this patch looks like prototype k." **CRAFT** [18] combined concept extraction with hierarchical activation factorisation, surfacing concepts at multiple abstraction levels in a single unified framework.

3 Explainability for Graph Neural Networks

GNNs dominate relational data tasks — molecular property prediction, social network analysis, knowledge graph reasoning — but standard feature-attribution methods designed for grids or sequences do not transfer because the natural explanation unit is a subgraph, not a scalar feature.

GNNExplainer [19] was the first general post-hoc method, identifying per-prediction subgraphs and node-feature subsets that maximise mutual information with the original prediction. **PGExplainer** [20] improved on this by training a separate network to predict edge masks, enabling generalisation across the test set rather than solving a per-instance optimisation. **SubgraphX** [21] applied Monte Carlo Tree Search to select subgraphs maximising a Shapley-value criterion, producing connected subgraphs that are more semantically coherent and easier for domain experts to interpret.

A recognised limitation across all three is the *faithfulness evaluation problem*: because the GNN's decision process is opaque, it is difficult to verify that an identified subgraph is genuinely causally responsible rather than merely correlated with the prediction.

4 Explainability in the Era of Transformers

4.1 The Attention Debate

The self-attention mechanism produces token-level weights that appear to explain which tokens influenced an output. Jain and Wallace [22] challenged this, showing attention weights often disagree with gradient-based importance scores and that adversarial attention patterns can change distributions without changing predictions. Wiegreffe and Pinter [23] gave a more nuanced response, showing attention *can* serve as faithful explanation under certain conditions. The community reached a working consensus: raw attention weights are a weak proxy; reliable attribution requires propagation through the full computation graph.

Abnar and Zuidema [24] introduced *attention rollout*, recursively multiplying attention matrices across layers. Chefer et al. [25] extended this with gradient signals weighted by LRP-derived relevance rules, consistently outperforming raw attention and rollout on faithfulness metrics. Probing work [26, 27] further showed that BERT's layers recover the classical NLP pipeline in order — lower layers encode part-of-speech, middle layers coreference, higher layers semantic roles — suggesting attention heads specialise functionally and that mechanistic analysis at the head level is tractable.

4.2 LLMs and Self-Explanation

The emergence of fluent LLMs created a new mode of explanation: prompting the model to produce a rationale for its own output. Wei et al. [28] showed that *chain-of-thought prompting* both explains the answer and contributes to producing it, improving performance on complex reasoning tasks. However, LLM-generated explanations can be factually incorrect or inconsistent with internal computation, motivating a sharp distinction between *plausible* explanations (coherent to humans) and *faithful* explanations (accurately describing the model's computation) [29].

4.3 Mechanistic Interpretability

Mechanistic interpretability seeks to reverse-engineer the algorithms implemented by neural networks at the level of individual weights and circuits. Elhage et al. [30] established a mathematical framework for analysing attention circuits in Transformers. Wang et al. [31] applied this to identify a specific five-layer circuit in GPT-2 Small responsible for indirect object identification. Conmy et al. [32] proposed automated circuit discovery (ACDC) to improve scalability. Bricken et al. [33] applied dictionary learning to decompose transformer MLP layers into sparse, monosemantic features, addressing the *superposition hypothesis* that networks represent more features than neurons by encoding them in overlapping directions.

While these case studies are impressive, circuits identified in small models do not obviously generalise to billion-parameter deployed models — a central open question for the programme.

5 Evaluation of Explanations

Evaluation is arguably the most consequential unsolved problem in XAI. Without rigorous evaluation, it is impossible to compare methods, assess progress, or trust a deployed explanation system.

Jacovi and Goldberg [29] formalised *faithfulness* for NLP, arguing it is the primary desideratum for explanations that audit or debug models, while *plausibility* (agreement with human intuition) matters for end-user explanations. These desiderata can conflict. **ROAR** (RemOve And Retrain) [34] operationalises faithfulness by removing features deemed important, retraining, and measuring the performance drop; it is expensive but more reliable than perturbation metrics that skip retraining. The **ERASER** benchmark [35] provides NLP datasets with both model-extracted and human-annotated rationales, enabling unified measurement of faithfulness and plausibility.

Human-centred evaluation adds further complexity. Lage et al. [36] found that simpler, more concise explanations improve decision speed and that explanation complexity interacts non-trivially with user expertise. Zhou et al. [37] identified three evaluation types — application-grounded, human-grounded, and functionally-grounded — which frequently disagree, underscoring the absence of a single authoritative paradigm.

6 Domain Applications

Healthcare. XAI faces the most intense deployment pressure in medicine. Tjoa and Guan [38] identified three consumer groups — clinicians, patients, and regulators — each with distinct requirements, and noted that saliency maps are often inconsistent and vulnerable to perturbations, making them unreliable for clinical use. Reyes et al. [39] argued that the key gap in radiology is workflow integration: explanation outputs must conform to radiological reporting lexicons, not just highlight pixels.

NLP. A distinctive challenge is that tokens are not semantically independent, so masking one token changes the meaning of surrounding context in ways that interact non-trivially with model behaviour [40].

Computer Vision. Concept-based and prototype methods have seen growing adoption for tasks requiring object-level explanations. ProtoPNet and its extensions consistently achieve competitive accuracy while being more robustly interpretable than post-hoc attribution on black-box models [17].

Autonomous Systems. Autonomous vehicles require explanations satisfying both technical (fault diagnosis, safety verification) and socio-legal (liability, certification) requirements. Atakishiyev et al. [41] identified a need for *contrastive temporal explanations* that account for counterfactual decisions not taken and the temporal context over which decisions unfold.

7 Emerging Challenges and Future Directions

Adversarial attacks on explanations. Slack et al. [42] demonstrated that a classifier can behave differently when queried by LIME or SHAP versus a real user, producing misleading explanations that conceal discriminatory behaviour. Heo et al. [43] showed adversarial perturbations can drastically alter an explanation while leaving the prediction unchanged. Together, these results imply that explanations cannot be trusted without meta-verification — itself an active research problem.

Privacy. Shokri et al. [44] showed that model explanations facilitate membership inference attacks, creating a fundamental tension in privacy-sensitive domains: the transparency regulators demand may expose the training data of the individuals regulation is meant to protect. Privacy- preserving XAI based on differential privacy remains an open problem with unsatisfactory utility-privacy tradeoffs.

Scalability to foundation models. Most XAI methods were developed on small models. Gradient-based methods are expensive and unclear under emergent behaviours; LIME and SHAP require many model evaluations, prohibitive for LLMs. Mechanistic interpretability offers a principled path, but manual circuit analysis and combinatorial explosion at scale present serious practical barriers.

Regulatory compliance. The EU AI Act [9], adopted in 2024, imposes transparency requirements on high-risk AI systems but does not precisely define what “meaningful information about the logic of automated decisions” requires. This creates substantial legal and technical ambiguity that demands collaboration between legal scholars, policymakers, and AI researchers.

Looking ahead, the most pressing research directions are: causal XAI grounded in structural causal models; scalable automated circuit discovery for large models; shared evaluation benchmarks analogous to ImageNet or GLUE; explanation methods tailored to generative models; and regulatory operationalisation of XAI requirements.

8 Conclusion

The period since 2019 has been transformative for XAI. The field has moved from a small set of feature-attribution methods applied to CNNs and tabular models to a rich ecosystem addressing a much wider range of architectures, tasks, and stakeholder needs. The attention debate, the rise of concept-based explanations, the emergence of mechanistic interpretability, GNN-specific methods, and increasing rigour in evaluation are all significant advances.

At the same time, the gap between methodological capability and deployment practice remains wide. Evaluation is inconsistent, explanations can be adversarially manipulated, privacy and transparency pull in opposite directions, and the transition to billion-parameter foundation models has outpaced existing methods. Rudin's challenge [8] — to prefer interpretable models over explained black boxes — resonates differently in the era of general-purpose foundation models, where a single opaque model powers thousands of downstream applications. Progress will require sustained attention not only to method development but to evaluation, human factors, causal reasoning, and the socio-technical contexts in which explanations are produced and consumed.

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